

Nizar Esmat Fawzy

Software Engineer
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PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience building and shipping production systems. Owns API design, PostgreSQL schema architecture, and integrations across 50+ REST endpoints and 5 client platforms in a 10+ engineer team. Strong CS fundamentals – data structures, algorithms, OOP, SOLID, system design – sharpened by ECPC competitive programming, plus TypeScript, Node.js, NestJS, Python/Django, and Socket.io.

EDUCATION

Nahda University <i>Bachelor of Computer Science</i>	Sep 2019 – Jul 2023 <i>Beni-Suef, Egypt</i>
Information Technology Institute (ITI) <i>Full Stack Development Track (Python)</i>	Nov 2024 – Mar 2025 <i>Smart Village, Egypt</i>
Completed a competitive full-stack training program: designed a normalized PostgreSQL schema for 10k+ users and built RESTful APIs in Python/Django with ETL-style data workflows, contributing to architecture and data modeling decisions.	
Certifications: Python Professional Track (ITI) SQL & PostgreSQL (Udemy) Node.js (Udemy) MERN Stack (Route Academy)	
Languages: Arabic (Native), English (Fluent)	

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

2nd Place – Nahda University ECPC Qualifiers (2021, 2022), competing in the Egyptian Collegiate Programming Contest;
Organizing Volunteer – ICPC/ECPC/ACPC, 2022–2024.
300+ algorithmic problems solved in C++ on Codeforces – data structures, algorithms, and complexity analysis.

PROFESSIONAL EXPERIENCE

Opream <i>Software Engineer</i>	May 2025 – Present <i>Cairo, Egypt</i>
- Design and ship 50+ RESTful API endpoints across 5 production platforms in TypeScript, NestJS, PostgreSQL, and React, owning each feature from API contract and data model design through testing and release. - Architected Mostaqbel Watan, a private social platform serving 1,000+ active users, with Socket.io private/group messaging, Jitsi video meetings, and an automated Facebook Graph API pipeline ingesting group posts. - Built Onway (ride-sharing, Fawry payment integration) and Al-Azhar & Bingo (unified lesson/library management); on Event Ticketing, eliminated inventory lock-up with a scheduled job releasing stale “pending” reservations every 10 minutes. - Improve query latency by normalizing schemas, adding targeted indexes, and eliminating N+1 patterns; cover critical paths with Jest unit and integration tests and review teammates’ pull requests to catch defects before release.	
iSchool <i>Software Instructor (Part-time)</i>	Jul 2024 – Nov 2024 <i>Cairo, Egypt</i>
Taught programming fundamentals and algorithmic problem solving, breaking down core CS concepts for students.	
WE School <i>Web Development Instructor</i>	Aug 2023 – Jun 2024 <i>Cairo, Egypt</i>
Taught web development with HTML, CSS, and JavaScript, guiding students from fundamentals to working projects.	

PROJECTS

ECPC Movement Control <i>NestJS, PostgreSQL, TypeORM, Socket.io, React, JWT</i>	GitHub
Built a real-time volunteer dispatch system for the Egyptian Collegiate Programming Contest: a scoring algorithm auto-assigns requests by hall proximity, gender match, and idle time, backed by a versioned TypeORM/PostgreSQL schema, an audit trail, and live Socket.io updates under JWT-secured roles.	
Chat Application <i>React, Node.js, Express.js, Socket.io, MongoDB, Cloudinary</i>	GitHub Postman
Engineered a full-stack real-time chat application (React/Zustand, Node/Express/Socket.io) with private messaging, group channels, and live presence; hardened the API with HTTP-only cookie JWT auth, Arcjet rate limiting, and Cloudinary-backed media storage with full CRUD.	

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++
Backend: Node.js, NestJS, Express.js, Django, REST APIs, Socket.io, TypeORM, JWT Authentication
Frontend: React, HTML, CSS
Databases: PostgreSQL, MongoDB, Data Modeling, Indexing, Query Optimization
Computer Science: Data Structures & Algorithms, Problem Solving, OOP, SOLID, Design Patterns
Testing & DevOps: Jest, Unit & Integration Testing, Git, GitHub, Docker, CI/CD, Linux
Languages: Arabic (Native), English (Fluent)